Introduction to plasma physics using Python

Introduction to Plasma Physics Using Python

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This book explores the fundamental physics of plasmas using python as a numerical gateway. The first volume deals with basic plasma physics, from single particle motion all the way to fluid dynamics and then MHD. It uses Jupyter notebooks to off-load numerical complexity onto a friendly interactive numerical interface. Students can focus on the concepts inherents to plasma physics rather than lengthy mathematical arguments. This book targets specifically senior undergraduate students and first year graduate students. It is actively used in a plasma physics course offered at the University of Rochester. This book will be kept as an online resource. Rapid changes in python libraries would make a printed version quickly obsolete. This project is suppored by the National Science Foundation under the grant nubmers PHY-1943939 and PHY-1725178.

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